



Environmental, Social, and Governance Signals and Investment Decision-Making: Evidence from Mutual Fund Analytics

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Abstract: The increasing integration of environmental, social, and governance (ESG) considerations into financial markets has raised fundamental questions regarding their roles in investment decision making. In particular, it remains unclear whether ESG-oriented investment strategies mirror substantive changes in portfolio construction or primarily follow prevailing market trends. This study examined the decision relevance of ESG signals by analyzing the behavior and performance of ESG-oriented mutual funds. Using a sample of 41 funds, a data-driven analytical framework was employed to evaluate portfolio composition, risk–return performance, and the determinants of ESG ratings. The analysis first considered whether ESG funds systematically allocated capital toward firms with stronger ESG profiles. Although a modest tilt toward higher-rated companies was observed, the differences relative to conventional funds remained limited and, in most cases, statistically insignificant, thus indicating that ESG considerations were not the dominant driver of portfolio selection. The second part evaluated fund performance within a risk–return framework using benchmark-adjusted measures and information ratios (IR). While all funds generated positive returns over the sample period, the majority failed to outperform their benchmarks. Only a small subset exhibited consistently favorable risk-adjusted performance. These findings suggested that ESG-oriented strategies did not provide a reliable basis for achieving superior financial outcomes and might involve trade-offs in portfolio allocation. Finally, cross-sectional regression analysis demonstrated that ESG ratings were strongly associated with firm-specific characteristics, especially size and profitability. This result indicated that ESG scores might reflect underlying financial capacity rather than deliberate sustainability-oriented decisions. Taken together, this study implied that ESG signals offered limited standalone values for guiding decisions of investment. Effective portfolio design therefore requires a broader analytical approach that integrates ESG metrics with conventional financial indicators.

Keywords: Environmental, social, and governance ratings; Investment decision; Mutual fund analytics; Portfolio allocation; Risk–return; Sustainable

1 Introduction

The integration of environmental, social, and governance (ESG) considerations into financial markets has increasingly influenced investment decision-making processes. Often viewed as an extension and evolution of the corporate social responsibility (CSR) framework and the stakeholder welfare maximization theory, ESG investing seeks to incorporate a broader set of considerations into the process of investment decision making. In doing so, it aims to move beyond the narrow focus of the traditional shareholder value maximization principle by accounting for the wider social, environmental, and governance impacts of corporate activity.

A closer examination of the concept reveals a number of controversies and unresolved issues. For this reason, this study addressed three analytical questions relevant to investment decision making. First, it examined whether the performance of ESG-oriented mutual funds differed significantly from that of conventional funds. Second, it investigated whether these funds' portfolios were genuinely tilted toward companies with high ESG performance. Finally, it explored whether high ESG ratings were primarily driven by firm-specific characteristics or reflected firms' deliberate efforts.

The ESG concept began to take shape in mid-2000s. It was first formally referenced in the 2004 report of the United Nations Global Compact and was further developed through initiatives such as the Principles for Responsible Investment (PRI), the Climate Disclosure Standards Board (CDSB), and the Sustainability Accounting Standards

Board (SASB). The core premise of ESG is that investors should adopt a more responsible approach to investment decisions by favoring companies whose activities contribute positively to environmental protection, social well-being, and high standards of corporate governance. Emerging from growing awareness of climate change and environmental degradation associated with business activity, increasing concern over social issues, and a series of corporate governance scandals in early 2000s, the ESG framework sought to introduce a broader and more human dimension into business and investment decision making. In doing so, it aims to move beyond the traditional emphasis on profits and profitability as the sole indicators of corporate success.

As noted above, the concept remains insufficiently defined across several dimensions, giving rise to numerous open questions. These include issues regarding the role and accountability of corporate managers who are expected to pursue multiple yet sometimes conflicting objectives; the relationship between the adoption of ESG principles and company performance; and the challenges associated with measuring inherently difficult-to-quantify factors such as environmental and social impacts. Nevertheless, the most fundamental dilemma arises from the very foundations of the concept itself, i.e., the clarity of its ultimate objectives and the appropriateness of the methods intended to achieve them.

The PRI, promoted by the United Nations, are defined as “a set of voluntary guidelines that aim to help investment institutions incorporate ESG factors into their decision-making and ownership practices”. Although this description may suggest that the initiative is primarily intended to secure a better ESG setting, the PRI does not explicitly frame its objective in those terms. Instead, the expected benefits of their implementation are described as “improved risk management”, “enhanced returns”, “increased engagement by investing companies in improving their ESG performance”, and “an enhanced reputation” for investors committing to responsible investment practices. While these outcomes may indirectly contribute to broader goals of global sustainability, they remain largely oriented toward improving investment performance within the traditional risk–return framework, thereby placing the broader societal objectives somewhat in the background.

Moreover, one of the fundamental events in the development and the broader adoption of ESG principles was the Business Roundtable (BRT) in 2019 which issued a statement that was signed by more than 93% of the Chief Executive Officers (CEOs) of the BRT member companies. It asserted that “while each of our individual companies served its own corporate purpose, we shared a fundamental commitment with all of our stakeholders”. As of today, at least 85% of these signatories published ESG reports as they are seen as a means of proving this broader societal commitment. However, the accompanying statements by many of the signees cast doubt on their own (and the broader) understanding of the redefined corporate purpose (e.g., “By taking a broader and complete view of corporate purpose, boards could focus on creating long-term value, better serving everyone—investors, employees, communities, suppliers, and customers” - Bill McNabb, former Chief Executive Officer (CEO) of Vanguard; or “. . . the best-run companies do more. They put the customer first and invest in their employees and communities. In the end, it’s the most promising way to build long-term value” - Tricia Grifit (President and CEO of Progressive Corporation). Obviously, there is another contradiction. It is unclear what the main objective of the “commitment to all stakeholders” is? Is it satisfaction of the needs and interests of all these stakeholders or is it merely a means of achieving the long-term goals for the shareholders, as the long-term value depends on it? We are well aware to whom the long-term value belongs.

In principle, there is nothing inherently problematic in pursuing stakeholders’ interests or adopting ESG principles. The concern arises with respect to the genuine commitment to the broader objectives associated with ESG, namely environmental protection, improvements in social and societal conditions, and the strengthening of corporate governance standards. There is a risk that these initiatives may, in some cases, amount to some kind of “window dressing”, whereby companies, their executives, investors, and other economic actors derive additional benefits primarily by aligning themselves with a prevailing trend rather than by implementing meaningful changes. If ESG initiatives are genuinely driven by altruistic intentions, one would expect a certain trade-off between their implementation and shareholders’ welfare. Nevertheless, such a trade-off is not explicitly acknowledged, as the pursuit of long-term value remains unaffected.

This study adopted a data-driven analytical perspective to evaluate the role of ESG signals in portfolio allocation and investment strategy. The basic goal of this paper is to explore the extent of real commitment to the implementation of the ESG principles in investing, as opposed to their acceptance as a subject of fashion. In this regard, several questions arise: What weight do the ESG-oriented mutual funds put on high ESG performers, compared with other companies? Is the ESG label only used to attract investors? What is the risk-adjusted performance of ESG mutual funds compared with their benchmarks? Do investors, represented by the mutual funds, sacrifice a part of their return for the sake of broader well-being? Is there a true commitment on the companies’ side in this field or are their high ESG scores pre-determined by certain company-related factors?

To this end, this research selected and analyzed 41 mutual funds with ESG orientation. The analysis was performed in three parts. First, this study analyzed the average ESG scores of the ESG mutual funds and the composition of their portfolios with respect to the ESG scores of the companies involved. The goal is to check if ESG is a real commitment, a real determination, or a trend that the followers as well as some business agents (appraisers) take advantage of?

The second part examined the efficiency of the funds based on their annual returns and information ratios (IR) and compared these results with appropriate benchmark indices. The purpose of this analysis was to evaluate the relative performance of ESG funds and to examine whether their investors might be willingly sacrificing a portion of their expected returns. Finally, a regression analysis of the companies included in these funds was conducted to determine the extent to which their ESG scores were influenced by specific characteristics at firm level, as opposed to the result of managerial initiatives in ESG dimensions.

The remainder of the paper is structured as follows. First, a comprehensive literature review will be provided to set the basis of the problems and elaborate the research hypotheses. The subsequent section presents the methodology applied for the empirical study of all the research hypotheses, as well as the sources of data. The fourth section consists of three subsections, each of which depicts the results obtained in the empirical analysis. The fifth section discusses the empirical findings, while the last part summarizes the conclusions of the study.

2 Literature Review

The integration of ESG principles into corporate objectives and their incorporation into finance have arguably represented one of the most debatable developments in corporate finance in recent decades. Despite extensive discussion, the role of ESG principles in supporting investment decision making remains ambiguous. Regardless of one's perspective, motivation, or academic focus, it is difficult to overlook the complexity and multidimensional nature of this issue when attempting to draw robust and enduring conclusions.

The expansion of ESG investing has reignited fundamental debates about corporate purpose, valuation, and the channels through which non-financial considerations affect asset prices. Friedman [1] argued that firms should maximize shareholder value subject to legal and ethical constraints, a view that underpins modern corporate finance. However, early empirical work in the field of corporate social performance (CSP) suggested a positive relationship between CSP and corporate financial performance (CFP), hence supporting stakeholder theory arguments [2]. However, replication evidence cast doubt on the robustness of this finding. While stakeholder-oriented perspectives argued that ESG engagement enhanced firm value through improved risk management, reputation, and long-term cash flows, critics cautioned that ESG might dilute managerial accountability and obscure value maximization. This perspective aligns with Jensen and Meckling's agency framework [3], which allows shareholders to have objectives beyond pure financial returns. Shareholders may derive utility from environmental quality or social outcomes, thus implying that firms could legitimately pursue ESG objectives if aligned with investors' welfare. Edmans [4] argued that ESG was neither revolutionary nor incompatible with shareholder value maximization. Applying standard tools from corporate finance and economics, he demonstrated that shareholder value, defined as the present value of all future cash flows, was inherently long term, and that ESG investments should be evaluated using traditional Net Present Value (NPV) logic. Rather than replacing shareholder value, ESG considerations could be integrated within existing economic frameworks, provided they generate either long-term cash flows or reflect shareholders' preferences over externalities.

One of the core premises of this study concerned whether investing in high-ESG-performing companies yielded different profitability compared with traditional portfolio management, which focused solely on optimizing the risk–return trade-off. For this reason, scholars have explored the channels through which corporate efforts in areas such as environmental protection, labor practices, managerial transparency, and accountability translate into higher firm valuations and, ultimately, improve shareholders' returns.

An important strand of related literature viewed ESG as a form of intangible capital that may not be fully reflected in stock prices. Edmans [5] provided early evidence that the market failed to fully value intangible assets such as employee satisfaction. Firms recognized as “Best Companies to Work for” earn significant long-run abnormal returns, suggesting that socially oriented practices could enhance firm value but are incorporated into prices only gradually. Khan et al. [6] introduced the concept of ESG materiality. Using industry-specific materiality classifications from SASB, they demonstrated that firms investing in “material” ESG issues significantly outperformed firms investing in immaterial issues. This finding reconciles prior mixed results by showing that ESG only enhances performance when aligned with financially material sustainability dimensions. Zhao and Murrell, while replicating a previous study, discovered a positive corporate social performance, i.e., corporate financial performance relationship [7]. Since it is often the case in research in economics and finance, they concluded that the results critically depended on the sample chosen, its size and structure, as well as the length of the period under analysis.

Firm-level studies further complicated the ESG-performance relationship. Abate et al. [8] documented a positive association between ESG performance and accounting outcomes, particularly for governance and environmental dimensions, while stock-return effects remained heterogeneous across regions and industries. Their findings revealed that ESG primarily enhanced resilience and cash-flow stability, rather than generating persistent abnormal returns. Mitsuyama and Shimizutani [9] examined the announcement of an “ESG Brand” in Japan and found no significantly abnormal stock price reaction around the event window. This suggested that positive ESG classification did not automatically translate into short-term market gains, especially when information was already anticipated or viewed

as non-material.

This evidence supports the notion that certain ESG-related investments, particularly those related to human capital, may improve long-term performance through productivity and stability of the workforce, even if short-term market reactions are mild. However, subsequent literature cautioned that such effects were not universal across ESG dimensions and might depend on visibility and verifiability. Edmans [4] emphasized a fundamental asset-pricing principle: If ESG characteristics were observable and priced, they should not produce abnormal returns. Sustainable firms may deliver higher cash flows, but if these expectations are already embedded in prices, realized returns need not exceed benchmarks.

Existing body of literature provided mixed and often conflicting evidence regarding the economic implications of ESG integration. While a number of studies suggested that ESG engagement might enhance firm value through improved risk management, reputation, or long-term cash flow stability, others found limited or no consistent impact on financial performance. Moreover, empirical results appeared to be highly sensitive to methodological choices, sample composition, and the specific ESG dimensions to be considered.

An additional limitation of the current literature is that it tends to examine ESG either at the firm or portfolio level, with relatively limited integration of these perspectives. As a result, the extent to which ESG signals translate into actual investment decisions, particularly in the context of portfolio construction and fund performance, remains insufficiently understood.

Although company financial performance is the basic indicator of its overall operating accomplishment, investors are primarily concerned with how this performance translates into market valuation and stock returns as only a distinct stream of the literature could explore this issue. Kastens [10] provided global panel evidence showing a dual effect: ESG ratings are negatively associated with short-term profitability, i.e., return on assets (ROA), but positively associated with market value. The study highlighted a trade-off where firms might sacrifice short-term earnings to achieve higher market valuations, particularly in developed markets and for large firms. This aligns with preference-based asset pricing theories to imply that investors reward ESG engagement through valuation premiums rather than realized returns.

A number of scholars took the opposite approach in exploring the ESG-valuation issue. Instead of studying the impact of positive ESG examples, they focused on poor ESG performers and the impact of ESG-related risks on company valuations. In an early study, Hong and Kacperczyk [11] introduced the term “sin stocks” to describe publicly traded companies operating in industries such as tobacco, alcohol, and gaming. Their findings showed that these stocks generated positive abnormal returns, which they interpreted as compensation for being excluded by norm-constrained investors. The authors linked these returns to the higher litigation risks associated with such firms and confirmed that this relationship held in markets beyond the United States.

Having revisited this evidence, Blitz and Fabozzi [12] demonstrated that “sin stock” outperformance largely disappeared once exposures to profitability and investment factors were controlled, so there was nothing special related to “sin stocks” that could not be compensated by investing in “non-sin stocks”. Their findings highlighted that return differentials associated with ESG exclusions might reflect systematic factor tilts rather than a distinct ESG premium; therefore, investors favoring high ESG performers should increase the weights of stocks that were exposed to factors that drove “sin stocks” returns. While extending this argument, Cornell and Damodaran [13] emphasized the distinction between “doing good” and “sounding good”. They argued that ESG affected valuation only insofar as it altered cash flows, growth, or risk, cautioning against embedding ESG narratives directly into valuation models without clear economic channels. ESG is unlikely to improve company operating performance, but the risk of “being bad” is more likely to increase its cost of capital. Therefore, ESG-oriented companies could provide higher returns to their shareholders, but these effects could only be available to early investors. Their work reinforces the view that ESG relevance is firm- and context-specific, rather than universal.

Given that investment fund behavior and performance are central to this study, it is important to examine the increasing attention among academics and practitioners to the relative performance of socially responsible funds, particularly ESG-oriented mutual funds, compared with their conventional counterparts. Results of these empirical studies, however, were far from being unambiguous. On the one hand, Lesser et al. [14] and Papathanasiou and Koutsokostas [15] concluded that the sustainability component in their investment strategies negatively affected fund performance. On the other hand, a series of more encouraging results were obtained by Gil-Bazo et al. [16], Alda [17], and Fang and Parida [18] in studies exploring both ESG and socially responsible investing (SRI) fund performance. Abate et al. [19], in an analysis carried out on a sample of 634 European mutual funds, concluded that funds investing in high ESG-rated securities had achieved superior efficiency compared with the general orientation benchmarks. Similarly, in a recent study, Hasnaoui [20] has obtained evidence of the superior efficiency of mutual funds with higher ESG ratings which “consistently outperform their lower-rated peers in both absolute and risk-adjusted returns”. He asserted that this could serve as an impetus for investors to pursue responsible investment strategies, particularly in the technology sector. However, there are numerous studies that reported no statistical difference in the performance between sustainable and conventional funds [21, 22]. Nofsinger and Varma [23] noted that the performance of ESG

funds was asymmetric. The funds were superior during market crises but potentially underperformed in markets during periods of stable equity.

A dominant concern among practitioners is whether ESG investing entails a financial sacrifice and to what extent these differences in performance can be attributed to the specific orientation of the funds, instead of the fund managers' abilities. Fish et al. [24] examined this question by constructing ESG-weighted portfolios using Bloomberg ESG scores across U.S. and European markets. Their findings suggested minimal differences in risk-adjusted returns between ESG-adjusted and conventional portfolios, indicating that investors need not sacrifice performance to pursue sustainability objectives. Nagy et al. [25] analyzed ESG tilt and ESG momentum strategies and revealed that ESG-based portfolios could outperform benchmarks over an eight-year period. However, only a part of the outperformance could be attributed to pure sustainability effects, while the remaining part was explained by factor exposures.

Finally, the study adopted ESG ratings from different providers and attempted to examine the extent of the impact of various rating providers on the selection of ESG-aware companies in individual and institutional portfolios. Evidence from the surveys of Amel-Zadeh and Serafeim [26] showed that institutional investors primarily used ESG information for risk management and engagement, rather than ethical screening alone. Investors also expressed concerns about data reliability, to reinforce the importance of measurement and uncertainty in ESG integration. Li and Polychronopoulos [27] extended this insight by demonstrating that portfolios constructed using ratings from different providers produced materially different compositions and performance outcomes. Even when applying identical portfolio construction rules, return dispersion across rating providers could be economically significant.

When taken together, the existing literature provided valuable insights into the relationship among ESG characteristics, firm performance, and investment outcomes. However, several important issues remain insufficiently resolved. First, it remained unclear to which extent ESG signals were consistently reflected in actual portfolio allocation decisions, as empirical evidence on the strength of ESG tilts in investment practice was mixed. Second, prior studies tended to focus on either firm-level ESG determinants or portfolio-level performance, with limited integration between these perspectives. Consequently, it was difficult to determine whether observed ESG outcomes were primarily driven by deliberate investment choices or by underlying firm characteristics.

Finally, the decision relevance of ESG metrics in practical investment settings has not been fully established. In particular, the role of ESG information in shaping risk–return trade-offs and guiding portfolio construction remains open to question, especially in the context of mutual fund strategies. These unresolved issues highlight the requirement for a more integrated analytical perspective that jointly examines portfolio behavior and firm-level ESG determinants.

3 Methodology and Data

Building on the preceding analysis, this study developed a data-driven framework to examine the role of ESG signals in portfolio allocation and investment performance. To ensure a clear linkage between the theoretical motivation and the empirical approach, the analysis was structured around three testable research hypotheses, each corresponding to a specific methodological framework.

Hypothesis 1: The portfolios constructed by ESG-oriented mutual funds differ significantly from those of their benchmark indices and comparable conventional funds.

Empirical approach: This hypothesis was tested by comparing the average ESG ratings of companies included in ESG fund portfolios with those of non-selected companies (from benchmark or general funds), using Analysis of Variance (ANOVA) and mean-difference tests.

Hypothesis 2: ESG-oriented funds do not exhibit systematic outperformance relative to benchmark indices in a risk–return framework.

Empirical approach: This hypothesis was examined through a comparative analysis of average returns, benchmark-adjusted performance (e.g., excess returns or IR), and risk–return scatterplots over the observed period.

Hypothesis 3: Firms' ESG ratings are significantly associated with their financial characteristics.

Empirical approach: This hypothesis was tested using cross-sectional regression models where ESG scores served as the dependent variable, and firm-level financial indicators, such as size, profitability, cash flow generation, etc. as explanatory variables.

With respect to the first hypothesis, for the analysis of the structure of the ESG fund portfolios, we examined 41 mutual fund portfolios that focused on investing in companies with a positive ESG impact (<https://www.mutualfunds.com/equity-categories/environmentally-responsible-equity-funds-and-etfs/>). The website provided information about a total of 51 mutual funds (as of February 2026), but the list was reduced to 41 funds for several reasons. The basic goal of the selection process was to provide sufficient representativeness for the various investment strategies pursued by the varieties of funds, and therefore, some of the funds required to be excluded. For some of the ESG funds, the parent (benchmark) fund was not clearly specified, while in other cases, the identification of comparable general-orientation benchmark funds was ambiguous due to the specialized investment strategies (e.g.,

dividend-focused or specific small-cap). Moreover, certain funds managed by the same provider followed highly similar investment policies and were therefore excluded to avoid duplication and redundancy in the results.

For all these funds, we identified their parent funds that provided the broader investment universe from which their portfolios were constructed. Since drawing reliable conclusions regarding individual investors' attitudes was nearly impossible without a direct survey, we derived indirect inferences by analyzing the composition of these funds' portfolios and by comparing their returns with those of their respective benchmarks. To achieve this purpose, all companies that belonged to the top 10 holdings of both ESG-oriented mutual funds and the respective parent (general orientation) mutual funds were extracted. The first group comprised 158 companies (drawn from 41 funds), while the second included 107 companies (from 23 funds). These two groups were not exclusive, as 64 companies belonged to both groups. ESG ratings for these firms were obtained from two sources: S&P Global Ratings (SP Global) and Morningstar Sustainalytics.

For the second hypothesis, the annual returns of all these funds for the period 2021–2025 were used to determine their overall efficiency. As regards the comparison with the benchmark indices, the element of risk was introduced to relate the returns to their variability, and the IR were calculated. The IRs are calculated as:

$$IR = \frac{\text{Portfolio return} - \text{Benchmark return}}{\text{Tracking Error}}$$

where,

IR = Information ratio

Portfolio return = Portfolio return for the period

Benchmark return = Return of the fund used as a benchmark

Tracking Error = Standard deviation of the difference between portfolio and benchmark returns

The benchmarks used in the analysis were the parent, base, or target indices defined in the respective fund profiles. These indices represent broader stock market benchmarks from which ESG funds select their constituent equities and whose performance the funds aim to track. Initially, the sample included 41 ESG funds; however, several funds existed for less than four years and were therefore excluded because the available time series was too short to calculate meaningful standard deviations. As a result, the final sample consisted of 37 funds. The period analyzed was characterized by very strong stock market performance, with the S&P 500 increasing by approximately 74% and the NASDAQ index rising by more than 100%.

In the last part of the empirical analysis, two cross-sectional regression models were applied in which ESG scores served as the dependent variable and firm characteristics act as explanatory and control variables. The models differed in terms of both the source of ESG ratings and the composition of the sample. The first regression was based on a sample of 82 firms selected from the top 100 companies ranked by ESG ratings by Investor's Business Daily (IBD) (www.investors.com), based on Dow Jones Newswire, a subsidiary of IBD. The sample was reduced from 100 to 82 firms due to limitations in the availability of financial data, which was obtained from *Yahoo Finance*.

The second regression included 120 companies representing the top 10 holdings of the ESG-oriented mutual funds analyzed in Sections 4.1. and 4.2. of this paper. This sample was likewise constrained by data availability, and ESG scores were sourced from S&P Global (www.spglobal.com). The two samples were not mutually exclusive as a number of firms appeared in both datasets.

The specifications of the regression and the list of variables are presented in the Section of empirical results.

4 Empirical Results

4.1 Empirical Analysis: Structure of Environmental, Social, and Governance Funds in Relation to General Orientation Funds

Hypothesis 1: The portfolios constructed by ESG-oriented mutual funds differ significantly from those of comparable conventional funds.

The average ESG scores of companies held in ESG-focused funds were calculated and compared to those held in non-ESG funds, in order to determine whether meaningful differences existed between the two groups. The findings were somewhat unexpected. Although companies in ESG funds exhibited slightly better average ESG scores than those in conventional funds, ANOVA testing indicated that these differences were not statistically significant. It is important to note that Morningstar Sustainalytics scores measure ESG risk, meaning that lower values reflect stronger ESG performance. The average ESG ratings are presented in Table 1.

The calculated mean and median figures implied that the average ESG scores of the companies in the ESG-oriented funds generally exceeded those of the companies in the portfolios of the general orientation funds. However, the differences were minimal, while the median values based on SP Global were indeed identical. ANOVA analysis was then adopted to test the relevance of these results, which are illustrated in Tables 2 and 3.

The ANOVA analysis further confirmed that the differences between the mean ESG ratings in both groups were not statistically significant, regardless of the source of ratings used. *P*-values higher than .05 indicate the null hypothesis

that there is no statistically significant difference between the means of the two groups and cannot be rejected, so it is concluded that although the overall scores of the ESG funds are higher, the difference should be considered marginal. The results suggested that while ESG funds did exhibit a tilt toward companies with higher ESG ratings, it remained debatable whether this orientation was strong enough to characterize them as genuinely committed to the underlying objectives and whether their willingness to sacrifice expected returns was limited. This finding directly addressed the central question posed in the title of the paper and provided grounds for rejecting Hypothesis 1.

Table 1. Average environmental, social, and governance scores and Analysis of Variance for the two groups of companies

Analyzed Group	Mean		Median	
	Morningstar	SP Global	Morningstar	SP Global
Companies in the ESG fund portfolios	18.67	46.50	17.37	44
Companies in the general orientation funds' portfolios	19.91	45.47	18.80	44

Note: ESG: environmental, social, and governance; SP Global: S&P Global Ratings.

Table 2. Analysis of Variance of both groups of companies—environmental, social, and governance from Morningstar Sustainability

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	98.59447	1	98.59447	2.070097	0.151415	3.877473
Within Groups	12383.26	260	47.62794			
Total	12481.86	261				

Note: SS: sum of squares; df: degrees of freedom; MS: mean squares; F: F ratio; P-value: probability value; F crit: F ratio criterion.

Table 3. Analysis of Variance of both groups of companies—environmental, social, and governance from S&P Global Ratings

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	73.23333	1	73.23333	0.290395	0.590432	3.877754
Within Groups	65063.76	258	252.1851			
Total	65137	259				

Note: SS: sum of squares; df: degrees of freedom; MS: mean squares; F: F ratio; P-value: probability value; F crit: F ratio criterion.

As noted earlier, 64 companies that appeared in the portfolios of general-orientation funds were also included in the ESG-focused portfolios. This fact was used to assess the consistency of the selection criterion, that is, whether ESG ratings served as the primary determinant for portfolio inclusion in ESG-oriented funds.

To examine this, the average ESG ratings for both groups of companies were calculated (those selected and those not selected by ESG funds) to determine whether a statistically meaningful difference existed between their mean ESG scores.

Table 4 below presents the average ESG ratings of companies included in ESG fund portfolios and those excluded from this group, based on the total sample of 107 companies drawn from general-orientation funds.

Table 4. Average environmental, social, and governance rating

Analyzed Group	Mean		Median	
	Morningstar	SP Global	Morningstar	SP Global
Companies included in the ESG funds (64)	18.78	45.89	17.25	42.00
Companies NOT included in the ESG funds (43)	21.60	44.80	20.60	44.50

Note: ESG: environmental, social, and governance; SP Global: S&P Global Ratings.

The results illustrated that the ESG-focused funds had tilted their portfolios toward higher presence of highly ESG-rated companies. However, it is observable that the differences in the average ESG ratings between the two groups were not very significant, and the median value in the S&P ratings for the companies selected by the ESG funds was actually lower than that of the not selected stocks.

The ANOVA results from Tables 5 and 6 highlighted that differences in average ESG scores between companies included in ESG-oriented fund portfolios and those that were not selected were statistically significant only when ESG ratings from Morningstar were used. This outcome provided further evidence supporting the credibility of mutual funds' inclination toward firms with stronger ESG profiles.

Table 5. Analysis of Variance of companies selected and not selected by environmental, social, and governance funds—environmental, social, and governance from Morningstar Sustainability

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	205.3675	1	205.367	4.385	0.039	3.932
Within Groups	4917.183	106	46.830			
Total	5122.551	107				

Note: *SS*: sum of squares; *df*: degrees of freedom; *MS*: mean squares; *F*: *F* ratio; *P*-value: probability value; *F* crit: *F* ratio criterion.

Table 6. Analysis of Variance of companies selected and not selected by environmental, social, and governance funds—environmental, social, and governance from S&P Global Ratings

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	29.45993	1	29.460	0.126	0.723	3.933
Within Groups	24048.67	106	233.482			
Total	24078.13	107				

Note: *SS*: sum of squares; *df*: degrees of freedom; *MS*: mean squares; *F*: *F* ratio; *P*-value: probability value; *F* crit: *F* ratio criterion.

The final step of the analysis involved comparing the ESG score–return profiles of all companies to evaluate whether ESG-oriented funds might have sacrificed part of their returns in pursuit of responsible investment objectives. For this purpose, we presented a graphical overview of the ESG risk–return profiles of the selected and non-selected companies. Figure 1 illustrates the distribution of firms in the “selected” group according to their ESG risk scores and their total realized five-year stock returns for the 2021–2025 period, while Figure 2 provides the same information for the non-selected group.

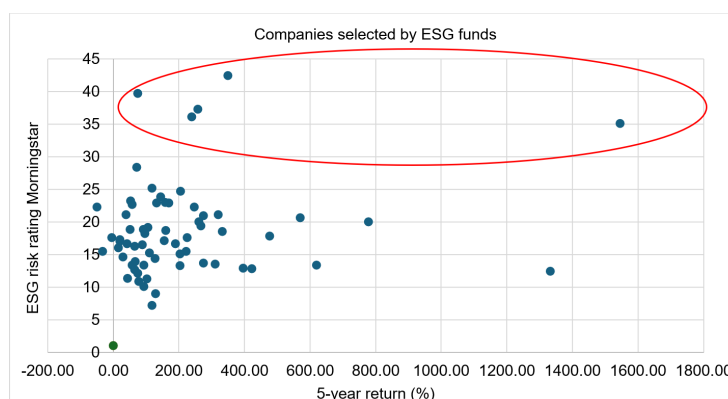


Figure 1. Environmental, social, and governance risk–stock return distribution of companies selected by environmental, social, and governance funds

Note: ESG: environmental, social, and governance.

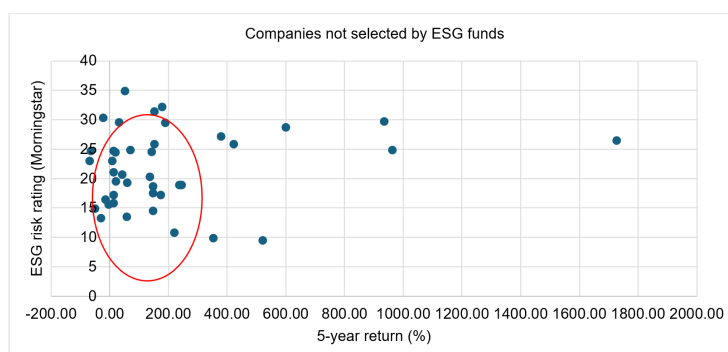


Figure 2. Environmental, social, and governance risk–stock return distribution of companies not selected by environmental, social, and governance funds

Note: ESG: environmental, social, and governance.

Figure 1 shows that most companies chosen by ESG-oriented funds exhibited ESG risk levels between 15 and 25.

However, several firms displayed substantially higher ESG risk levels (highlighted with a red circle) were likely to be included due to expectations of superior returns, an expectation that appeared to have materialized in at least one case.

Figure 2 reveals that several companies with comparatively low ESG risk scores were not included in ESG funds' portfolios (also marked with a red circle). Taken together, these observations suggested that ESG ratings were not the sole determinant in constructing ESG-focused portfolios. While it is reasonable to assume that fund managers would not disregard other important criteria, particularly expected returns, the overall impression, supported by the mean comparisons and ANOVA results, indicates that further research is warranted to assess the extent to which the ESG designation in these funds' names is substantiated.

The final section of this part presents an analysis of the average ESG rating of the mutual funds under consideration. This assessment was based on the ESG scores of the companies included in each fund's top 10 holdings, weighted in accordance with their respective shares in the fund's portfolio.

Although the top 10 holdings did not fully reflect the entire portfolio composition, they could still provide a reasonable approximation of fund managers' attitudes toward the trade-off between ESG performance and fund returns. Accordingly, the ESG scores of the companies in the top 10 holdings of both ESG-oriented funds and general-orientation funds were weighted by their portfolio shares and further adjusted to account for the proportion of the top 10 holdings within the fund's total portfolios.

The results of this analysis are summarized in Table 7.

Table 7. Average environmental, social, and governance scores per group of funds

Group of Funds	Morningstar		SP Global	
	ESG Funds	Non-ESG Funds	ESG Funds	Non-ESG Funds
International/global/emerging markets	16.91 (14)*	18.84 (6)	59.87 (14)*	55.29 (6)
US stocks	15.84 (9)*	17.24 (6)	45.56 (9)*	45.45 (6)
Large-cap	16.22 (5)*	21.81 (1)	47.33 (5)*	33.73 (1)
Small- and mid-cap	21.24 (5)*	22.24 (3)	34.94 (5)*	34.11 (3)
Growth stocks	17.06 (4)	16.47 (3)*	43.58 (4)	43.57 (3)

Note: The numbers in brackets represent the number of funds in the particular group. The total number of funds does not add up to the total number of funds analyzed, as some of them are classified into two or more groups (for example, Large-cap US equity fund was classified as both US-oriented and large-cap). The asterisk denotes the higher score for the particular group and ESG score provider. ESG: environmental, social, and governance; SP Global: S&P Global Ratings.

Although the results presented above were not based on comprehensive samples, they allowed several meaningful observations. In the case of both global- and US-focused funds, the average scores of the ESG funds were higher than those of the non-ESG group, which was consistent with the expectations. However, the comparison between the large-cap and small- and mid-cap funds showed that these differences were substantially higher in the former group compared with the latter. While acknowledging that these differences largely reflected the investment strategies of individual fund managers, the relatively smaller gap within the small- and mid-cap group indicated that the choice of the managers within this group was more limited.

Furthermore, the size of the difference between the large-cap ESG-funds and small- and mid-cap ESG-funds observed across both ESG metrics indicated that smaller companies were less likely to achieve higher ESG scores. This finding is consistent with the regression results in Section 4.3, confirming the positive association between company size and ESG scores. A similar conclusion could be reached with the results of the growth stock funds, implying that companies in the stages of faster growth are less likely to dedicate considerable resources to their ESG-related efforts.

4.2 Empirical Analysis: Analysis of the Efficiency of Environmental, Social, and Governance Funds

Hypothesis 2: ESG-oriented funds do not systematically outperform their benchmark indices in terms of risk and return.

This section examined the efficiency of ESG funds within a risk–return framework, both on a stand-alone basis and relative to their benchmark funds. The first objective is to evaluate the absolute returns generated by ESG funds over the past five years and to gain a general understanding of their capacity to satisfy investors apart from the non-pecuniary benefits they may offer. The second objective is to compare these returns with those of their parent indices used as benchmarks on the basis of having similar investing strategies, in order to determine the effect that constructing portfolios by selecting companies with higher ESG scores has on their risk–return performance.

The analysis of ESG funds indicated that 37 funds generated positive average annual returns during the observed period. However, assessing only their absolute profitability was insufficient; it is also necessary to evaluate their performance relative to their benchmarks. To address this, the differences between the average annual returns of the ESG funds and those of their respective benchmarks were calculated. The results were striking: 27 out of the 37 ESG funds recorded average returns that lagged behind their benchmark indices.

The average returns, excess returns, and the IR for the funds analyzed are given in Table 8.

Table 8. Average returns and excess returns over the benchmark and information ratios of environmental, social, and governance funds

No.	ESG Fund	Average Annual Return	Average Annual Excess Return (In Percentage Points)	IR
1	Boston Common ESG Impact International Fund	3.80%	-6.22	-3.86
2	Boston Common ESG Impact US Equity Fund	14.06%	-1.90	-1.10
3	ClearBridge Large Cap Growth ESG ETF	14.74%	-0.41	-0.04
4	Fidelity International Sustainability Index Fund	7.78%	-1.29	-1.04
5	Fidelity US Sustainability Index Fund	16.54%	0.32	1.80
6	FlexShares ESG & Climate US Large Cap Core Index Fund	11.08%	-1.70	-1.75
7	FlexShares STOXX Global ESG Select Index Fund	13.84%	1.81	0.17
8	FlexShares STOXX US ESG Select Index Fund	14.54%	-1.42	-0.67
9	Hennessy Stance ESG ETF	4.40%	-8.37	-1.00
10	Invesco ESG NASDAQ 100 ETF	17.83%	1.35	3.08
11	Invesco ESG NASDAQ Next Gen 100 ETF	5.55%	0.92	0.37
12	iShares ESG Advanced MSCI EAFE ETF	7.68%	-2.34	-0.57
13	iShares ESG Advanced MSCI EM ETF	5.90%	0.25	0.06
14	iShares ESG Advanced MSCI USA ETF	16.16%	1.16	0.52
15	iShares ESG Aware MSCI EAFE ETF	9.78%	-0.24	-0.30
16	iShares ESG Aware MSCI EM ETF	5.36%	-0.29	-0.20
17	iShares ESG Aware MSCI USA ETF	14.72%	-0.28	-0.71
18	iShares ESG MSCI EM Leaders ETF	4.06%	-1.59	-1.23
19	iShares ESG MSCI KLD 400 ETF	15.70%	1.34	0.49
20	iShares ESG Optimized MSCI USA ETF	14.20%	-0.80	-0.33
21	iShares ESG Optimized MSCI USA Min Vol Factor ETF	4.88%	-0.67	-0.31
22	iShares ESG Select Screened S&P 500 ETF	16.18%	2.05	0.74
23	iShares ESG Select Screened S&P Mid-Cap ETF	9.24%	-1.98	-0.21
24	iShares ESG Select Screened S&P Small-Cap ETF	7.68%	-1.37	-0.16
25	Nuveen ESG Dividend ETF	6.80%	-5.33	-0.50
26	Nuveen ESG Emerging Markets Equity ETF	4.92%	-0.73	-0.24
27	Nuveen ESG International Developed Markets Equity ETF	9.66%	-0.36	-0.31
28	Nuveen ESG Large-Cap ETF	12.64%	-2.36	-1.09
29	Nuveen ESG Large-Cap Growth ETF	16.08%	-3.36	-0.58
30	Nuveen ESG Large-Cap Value ETF	9.84%	-1.00	-0.43
31	Nuveen ESG Mid-Cap Growth ETF	3.52%	-5.33	-1.74
32	Nuveen ESG Mid-Cap Value ETF	10.22%	0.91	0.34
33	Nuveen Winslow Large-Cap Growth ESG ETF	13.63%	1.31	0.12
34	State Street® SPDR® S&P 500® ESG ETF	16.82%	-0.92	-0.07
35	Vanguard ESG International Stock ETF	8.16%	-1.40	-1.52
36	Vanguard ESG US Stock ETF	14.92%	-0.02	-0.27
37	Xtrackers MSCI EAFE Selection Equity ETF	8.14%	-0.05	-0.29

Note: ESG: environmental, social, and governance; IR: information ratios.

What can be inferred from the above results? The commonly accepted threshold for the IR was around 0.5, which suggested that only five of the analyzed funds consistently outperformed their respective benchmarks. Although the observed period was relatively short, the results provided some indications of the efficiency of ESG funds. It was also possible that the weaker performance of these funds relative to their benchmarks reflected the abilities of their managers in stock selection. The findings did not allow a definitive conclusion that investing in ESG funds or ESG-oriented stocks necessarily led to lower returns. However, they highlighted the possibility that ESG-conscious investing might involve sacrificing at least a portion of the potential financial return.

4.3 Empirical Analysis: Environmental, Social, and Governance Ratings Dependent on Company Characteristics

Hypothesis 3: Firms' ESG ratings are significantly associated with their financial characteristics.

In this section, we examined the hypothesis that ESG performance was strongly influenced by firm-specific characteristics such as company size, profitability, leverage, and cash flows. This assumption rested on two key arguments. First, larger and more financially efficient firms are generally better positioned to allocate resources to activities that may be perceived as non-core business, including environmental initiatives and improvements in labor conditions. Second, firms that are larger and more profitable are typically at more advanced stages of their corporate life cycles, hence suggesting that they may have already attained higher standards in ESG practices.

Prior research documented significant relationships between ESG performance and corporate social responsibility on one side, and firm profitability [28], firm size [29], and financial constraints [30, 31] on the other. A crucial consideration, however, is that ESG scores may not perfectly capture a firm's genuine commitment to improving ESG

standards. Rather, they may reflect a company's capacity to achieve such improvements or, in certain instances, its ability to project a more favorable public image.

As mentioned above, two cross-sectional regression models were estimated. Company ESG scores were used as dependent variables, while the firm characteristics served as explanatory and control variables. The first regression was based on a sample of 82 firms selected from the top 100 companies ranked by ESG ratings by IBD, while the second regression included 120 companies, representing the top 10 holdings of the ESG-oriented mutual funds analyzed above.

The financial variables collected included measures of firm size, profitability, leverage, and operating cash flows, consistent with the determinants identified in prior literature. The financial dataset covered the most recent available period from 2022–2024. Given the annual frequency of ESG ratings, yearly data was used. However, the analysis was conducted as a cross-sectional regression rather than a panel regression, employing observations from either the most recent year (2024) or the average values over the sample period. Furthermore, in the first regression, industry dummies were used to elicit the probable impact of particular industrial sectors on company ESG results.

In addition to these variables in previous studies, the rationale for their inclusion in the regression model could be easily explained on the basis of their accounting connotation and business significance. The *size* of the company was considered a relevant determinant, as it was reasonable to assume that larger firms were better equipped to undertake ESG-related initiatives, given their greater access to both financial and human resources. A similar argument applied to *profitability* as a factor, as more profitable companies were assumed to have sufficient funds to dedicate to goals other than increasing owners' welfare, while firms facing challenges to profitability were less likely to allocate time and resources to activities perceived as non-essential. Even when profitability levels were satisfactory, the availability of liquid assets could create a constraint in pursuing different company activities. Therefore, *availability of cash*, particularly when generated from ongoing operating activities, was fundamental for the implementation of various strategic initiatives. Consequently, its role in attaining higher ESG ratings warranted careful examination, so the amount of cash held by firms was included as a separate variable in the model.

Conversely, higher *leverage*, measured by the relative amount of company debt, might constrain a firm's ability to engage in ESG efforts, since meeting debt obligations typically took precedence to ensure business continuity. The relevance of *profitability growth* was more ambiguous. On the one hand, firms experiencing rising profitability might be more inclined to invest in ESG activities; on the other hand, such firms might simultaneously require substantial financial resources to sustain their growth trajectory. Finally, certain *industries* were inherently placed to adopt environmentally friendly practices or enhance working conditions due to the nature of their operations. In this light, this paper controlled the industry effects when analyzing ESG ratings. Four sector groups were defined and they were (1) IT, computer equipment, and electronics; (2) Finance; (3) Construction, construction products, and materials; and (4) Other industries.

The inclusion of all these variables as explanatory, with the ESG level as a dependent variable in a cross-sectional regression model was necessary to reveal their roles in determining the levels of ESG, as assessed by the appropriate vendor of ESG ratings.

The basic regression model is:

$$ESG_i = \beta_0 + \beta_1 SIZE_i + \beta_2 PROF_i + \beta_3 CFO_i + \beta_4 LEV_i + \beta_5 GROWTH_i + \gamma Industry_I + \varepsilon_i \quad (1)$$

where,

- ESG_i denotes the ESG rating of firm i , obtained from Investors Business Daily (source of ESG ratings: MSCI ESG Research) and from S&P Global in the alternative specification. ESG values are transformed into their natural logarithmic form;

- $SIZE_i$ represents the size of company i , measured by the log value of either total assets or total revenues;

- $PROF_i$ is the relative profitability of company i , proxied by the ROA ratio, calculated as the ratio of either net income or operating income to total assets. Given the volatility of this measure, we used the three-year average of ROA over the sample period;

- CFO_i is the ratio of cash flow from operating activities (CFO) to total assets. Given the substantial volatility of CFO, we used the three-year average of CFO as in the case of ROA;

- LEV_i measures the leverage of company i , calculated as a ratio of long-term debt to total assets;

- $GROWTH_i$ represents the growth rate of profitability of company i , computed as the difference between the log value of operating income or net-income in 2024 and the corresponding value for 2022;

- $Industry$ is a categorical dummy variable indicating the industrial sector in which the firm operates. This variable was used only in Regression 1 (the 82-firm sample), as sectoral classifications were readily available for these companies.

Two regression specifications were estimated:

Regression 1: Based on a sample of 82 firms selected from the top 100 ESG-ranked companies by Investors Business Daily. Industry dummy variables were included.

Regression 2: Based on a sample of 120 firms drawn from ESG-oriented mutual fund portfolios analyzed in Section 4.1. ESG scores were sourced from S&P Global. Industry dummies were excluded due to limitations of the data.

As correlation analysis determined that the ROA and CFO variables were highly mutually correlated, they were never used jointly in the same regression specification.

Table 9 contains the results of Regression 1 whereas Table 10 displays the results of the second regression with the 120-company sample. The regressions were solved using Ordinary Least Squares (OLS) technique.

Table 9. Results of regression 1 with environmental, social, and governance (Dow Jones Newswire) as a dependent variable

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Total assets	0.026*** (0.000)		0.024*** (0.001)		0.027*** (0.000)			0.028*** (0.001)	
Total revenues		0.023*** (0.001)		0.021*** (0.002)		0.023*** (0.002)	0.020** (0.007)		0.025*** (0.003)
ROA (net income)	0.498*** (0.001)	0.443*** (0.004)			0.528*** (0.001)	0.455** (0.005)			
Net income growth	0.019 (0.557)	0.020 (0.540)	-0.011 (0.752)	-0.007 (0.833)	0.025 (0.465)	0.023 (0.501)	-0.006 (0.865)		
ROA (operating income)								0.343** (0.008)	0.230** (0.023)
Operating income growth								0.012 (0.679)	0.005 (0.875)
Cash flow - operating activities			0.358*** (0.001)	0.331*** (0.004)			0.351*** (0.004)		0.146 (0.554)
Leverage	-0.006 (0.859)	-0.003 (0.942)		-0.010 (0.840)	0.023 (0.667)	0.027 (0.614)	0.022 (0.691)	-0.025 (0.614)	0.014 (0.785)
ICT and electronics					-0.068 (0.289)	-0.097 (0.181)	-0.102 (0.157)		
Finance					-0.074 (0.263)	-0.085 (0.207)	-0.092 (0.173)		
Construction and materials					-0.066 (0.392)	-0.111 (0.151)	-0.108 (0.159)		
Other sectors					-0.060 (0.425)	-0.106 (0.164)	-0.117 (0.123)		
Intercept (C)	3.4288 (0.000)	3.512 (0.000)	3.483 (0.000)	3.564 (0.000)	3.458 (0.000)	3.613 (0.000)	3.693 (0.000)	3.388 (0.000)	3.462 (0.000)
Included observations	81	76	76	76	76	76	76	75	75
R^2	0.313	0.283	0.312	0.283	0.331	0.305	0.309	0.286	0.3155
DW statistics	2.102	2.177	2.169	2.210	2.116	2.145	2.186	1.953	2.017

Note: P -values in parentheses. *, **, and *** denote significance at 10%, 5%, and 1% respectively. ROA: return on assets.

Table 10. Results of regression 2 with environmental, social, and governance (S&P Global) as a dependent variable

Variable	(1)	(2)	(3)	(4)	(5)	(6)
Total assets	0.037** (0.012)	0.038** (0.006)	0.037** (0.011)			
Total revenues				0.040** (0.010)	0.034** (0.019)	0.041*** (0.008)
ROA (net income)	-0.156 (0.824)			-0.455 (0.506)		
ROA (operating income)		-0.422 (0.473)			-0.573 (0.332)	
Cash flow from operating activities			-0.076 (0.885)			-0.378 (0.763)
Net income growth	0.010 (0.779)		0.012 (0.738)	0.003 (0.940)		0.008 (0.813)
Operating income growth		-0.018 (0.705)			0.020 (0.683)	
Leverage	-0.070 (0.802)	0.035 (0.899)	-0.074 (0.793)	-0.062 (0.823)	0.061 (0.832)	-0.062 (0.824)
Intercept (C)	2.915 (0.000)	2.912 (0.000)	2.905 (0.000)	2.888 (0.000)	3.039 (0.000)	2.876 (0.000)
Included observations	83	80	84	83	80	83
R^2	0.099	0.115	0.099	0.104	0.092	0.105
DW statistics	1.860	2.097	1.864	1.845	2.073	1.870

Note: P -values in parentheses. *, **, and *** denote significance at 10%, 5%, and 1% respectively. ROA: return on assets.

Although only a limited number of coefficients were statistically significant, the regression results yielded several

noteworthy insights. Across all model specifications, firm size remained consistently significant, irrespective of how it was measured. As anticipated, larger firms demonstrated higher ESG efficiency, thus supporting our expectation that engagement in ESG activities was largely shaped by a company's underlying conditions and available resources, at least as much as by managerial commitment. This is valid even in specifications in which we control for the industrial sector.

5 Discussions

Discussion of the results is presented in accordance with the structure of the empirical analysis.

5.1 Environmental, Social, and Governance Fund–Portfolio Structure

The first research hypothesis, which assumed that ESG-oriented mutual funds constructed portfolios that differed significantly from those of comparable conventional funds, was not supported by the empirical findings. The analysis of average ESG scores and portfolio composition illustrated that companies included in ESG-oriented funds did not exhibit substantially stronger ESG profiles than those observed in general-orientation funds. In addition, the ESG characteristics of firms selected by ESG funds did not differ significantly from those of firms excluded from their portfolios.

These results called into question the extent to which ESG considerations were systematically incorporated into portfolio construction. Although a modest tilt toward higher ESG-rated firms could be observed, the magnitude of this effect remained limited and did not indicate a consistent or dominant role of ESG criteria in allocation decisions.

The graphical analysis of the relationship between ESG scores and stock returns further supports this interpretation. The absence of a clear and consistent selection pattern suggests that ESG considerations were applied in conjunction with, rather than in place of, traditional investment criteria. In particular, the inclusion of firms with relatively higher ESG risk alongside firms with stronger ESG profiles indicates that expected financial performance continues to play a central role in portfolio selection.

Taken together, the evidence suggests that ESG criteria are not systematically prioritized in the construction of ESG-oriented portfolios. Instead, they appear to function as one component within a broader decision framework, in which conventional return considerations remain dominant.

5.2 Efficiency of Environmental, Social, and Governance Funds

The second hypothesis, which stated that ESG-oriented funds did not systematically outperform their benchmark indices, could not be rejected. Empirical analysis indicated that the majority of ESG funds underperformed their respective benchmarks, while only a small number achieved consistently favorable IR.

Although these findings should be interpreted with caution due to the size of the sample and the specific characteristics of the observed period, they did not provide evidence supporting superior financial performance associated with ESG-oriented investment strategies. This outcome is consistent with the possibility that ESG-related characteristics are already reflected in asset prices, thereby limiting their potential to generate excess returns.

At the same time, the results might reflect underlying trade-offs between sustainability objectives and financial performance. It is possible that companies with higher ESG-related risks offer higher expected returns, or that ESG-related information is incorporated into valuation in advance, leading to comparatively lower realized returns.

From an investment perspective, these findings suggested that ESG considerations alone did not provide a sufficient basis for portfolio optimization. Rather, they should be evaluated alongside traditional financial indicators when assessing risk–return trade-offs and making allocation decisions.

5.3 Determinants of Environmental, Social, and Governance Scores

The third hypothesis, which proposed that firms' ESG ratings were associated with their financial characteristics, was supported by the regression results. Firm size emerged as the most consistent determinant across model specifications, while profitability and availability of cash flow exhibited explanatory power in certain cases.

These findings indicated that ESG performance was closely linked to the underlying economic capacity of firms. Larger and more profitable companies were more likely to allocate resources to ESG-related activities, hence suggesting that higher ESG scores might reflect structural advantages rather than solely the outcome of deliberate sustainability-oriented strategies.

The results also implied that ESG ratings could not be interpreted independently of firm characteristics. The observed relationships made it difficult to distinguish whether ESG outcomes primarily reflected managerial commitment or arose as a consequence of firm size, profitability, and resource availability.

Finally, the consistency of the results across alternative specifications strengthens their robustness, particularly with respect to firm size. Meanwhile, the differences observed across ESG rating providers highlight the limitations associated with the measurement of ESG performance, thus reflecting variations in underlying methodologies and evaluation criteria.

6 Conclusions

This study examined whether ESG-oriented investing reflected a substantive shift in investment behavior or largely represented a reconfiguration of conventional investment practices under a sustainability-oriented framework. By jointly analyzing portfolio composition, fund performance, and firm-level determinants of ESG ratings, the findings supported a more cautious interpretation of the role of ESG in investment decision-making.

First, the analysis of portfolio composition indicated that ESG-oriented funds exhibited only limited differentiation from their conventional counterparts. Although some evidence of a tilt toward higher ESG-rated firms was observed, this effect was neither strong nor consistent across the measurement. The sensitivity of results to ESG data providers further underscored the lack of standardization in ESG metrics, suggesting that conclusions regarding the ESG commitment of funds remained contingent on measurement choices.

Second, the empirical results did not provide evidence of systematic outperformance associated with ESG-oriented investment strategies. While all analyzed funds generated positive returns, most failed to exceed their respective benchmarks, and only a small number achieved stable risk-adjusted outperformance. These findings are consistent with the view that ESG-related characteristics are either already incorporated into asset prices or associated with underlying risk–return trade-offs, thereby limiting their capacity to generate excess returns.

Third, the analysis of firm-level determinants revealed that ESG ratings were closely associated with conventional financial characteristics, particularly firm size and profitability. This suggested that ESG outcomes were, to a significant extent, shaped by firms' economic capacities and structural attributes. As a result, higher ESG scores may not necessarily reflect stronger sustainability-oriented commitment, but rather the ability of firms to meet evaluation criteria.

To sum up, these findings indicated that ESG signals had limited standalone explanatory power in guiding investment decisions. Instead, they should be interpreted within a broader analytical framework that incorporates both financial fundamentals and firm characteristics when evaluating portfolio allocation strategies.

Several implications follow from these results. First, the objectives of ESG-oriented investing require clearer conceptual definition, particularly with respect to the relationship between sustainability goals and financial performance. Second, the observed disconnect between ESG indicators and measurable outcomes suggests that reliance on voluntary adoption mechanisms may be insufficient to ensure meaningful impact. Third, the lack of consistency across ESG rating methodologies highlights the need for improved measurement frameworks to enhance comparability and reliability. Last but not least, greater transparency and accountability in ESG evaluation processes are necessary to prevent the dilution of ESG principles into purely reputational or commercial instruments.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

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